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Beyond Space and Time

Holography, Black Holes, and the Limits of Locality

Before Stephen Hawking's death in 2018, he was the most famous living scientist in the world by a comfortable margin. That celebrity was entirely deserved; not only was Hawking a charismatic and influential public figure, and not only did he have an inspirational personal story, but his scientific contributions were incredibly significant in their own right.

Hawking's greatest achievement was showing that, once we include the effects of quantum mechanics, black holes "ain't so black," as he liked to say. Black holes actually emit a steady stream of particles out into space, and those particles carry energy away from the black hole, causing it to shrink in size. This realization led both to profound insights (black holes have entropy) as well as unexpected puzzles (where does the information go when black holes form and then evaporate away?).

The fact that black holes radiate, and the implications of that surprising idea, are the single best clue we have about the nature of quantum gravity. Hawking didn't first construct a full theory of quantum gravity and then use it to show that black holes radiate. Instead, he used a reasonable approximation, treating spacetime itself as classical, with dynamical quantum fields living on top of it. We hope that this is a reasonable approximation, anyway; but some of the puzzling aspects of Hawking's insight have given us second thoughts. Forty-five years after Hawking's original paper on the subject, trying to understand black-hole radiation is still one of the hottest topics in contemporary theoretical physics.